

Evaluation of Journal Policies to Increase Promotion of Transparency and Openness in Sport Science Research



Increased transparency and openness of sport science research will improve the ability to appraise, replicate, and implement research findings. However, sport science journals need to support and prioritize practices that bolster transparency and openness to facilitate this.^{1,2} Journal policies that actively support transparency and openness have been shown to improve the quality, reproducibility, and replicability of research.³ To date, over 1100 journals and funding agencies have ascribed to implementing 1 or more of the 8 standards for transparent and open research described in the Transparency and Openness Promotion (TOP) guidelines.⁴

We report an appraisal of how well the policies of the leading 38 sport science journals support transparent and open research practices. *Arthroscopy* was evaluated using the TOP Factor (<https://osf.io/t2yu5/>)—an alternative metric for evaluating the degree to which journal policies promote transparency and openness⁵—and the International Committee of Medical Journal Editors (ICMJE) disclosure-of-interest requirements.⁶ The TOP Factor is calculated as the sum of journal implementation of the 8 modular standards within the TOP guidelines and an additional standard related to publication of replication studies. The requirements for disclosure of conflicts of interest were evaluated using the 4 standards stated on the ICMJE disclosure form.⁶ *Arthroscopy*'s TOP Factor and conflict-of-interest scores are provided in Figures 1 and 2, respectively, alongside summary data from the top 38 sport science journals.

In brief, *Arthroscopy* scored well for TOP Factor item “study preregistration”; scored poorly for items “data citation,” “code citation,” “code transparency,” “data transparency,” “research materials transparency,” “analysis preregistration,” and “replication”; and had a total TOP Factor score of 3 of 27 (a higher score represents increased an requirement for transparency and openness by the journal). *Arthroscopy* was above the

mean TOP Factor score of 2.05 ± 1.99 of 27 and was above the mean ICMJE disclosure-of-conflict score (2.84 ± 1.62 of 4) compared with the top 38 sport science journals.

We strongly encourage *Arthroscopy* to review and, where necessary, update the journal's policies (<https://cos.io/top/> provides advice and resources to implement this). Greater transparency and openness will improve the conduct and translation of sport science research to ultimately move the field forward. Moreover, complete and transparent disclosure of conflicts of interest will increase confidence in research findings. By publishing this letter, *Arthroscopy* is taking action toward improving the transparency and openness of sport science research.

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